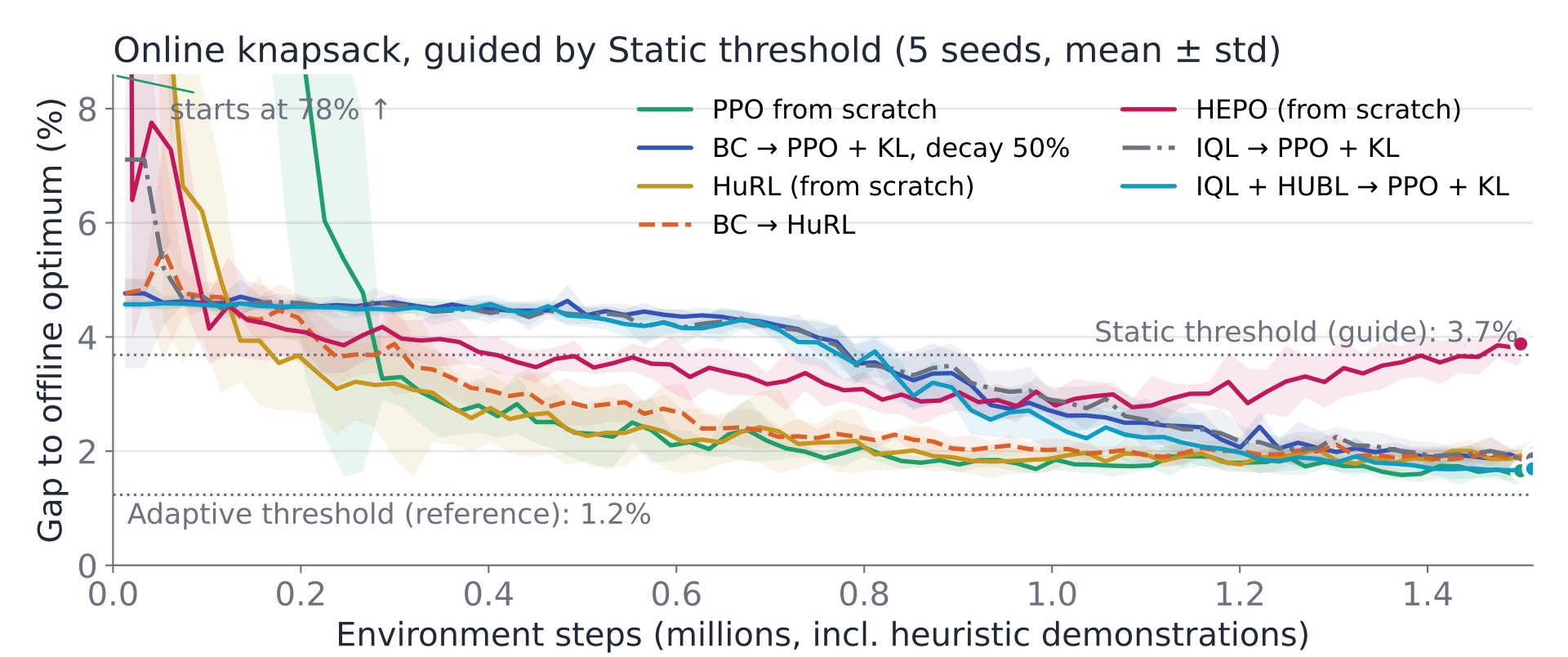
Online knapsack, guided by Static threshold (5 seeds, mean ± std)
starts at 78% ↑
PPO from scratch
BC → PPO + KL, decay 50%
HuRL (from scratch)
BC → HuRL
HEPO (from scratch)
IQL → PPO + KL
IQL + HUBL → PPO + KL
Static threshold (guide): 3.7%
Adaptive threshold (reference): 1.2%
Gap to offline optimum (%)
0
2
4
6
8
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Environment steps (millions, incl. heuristic demonstrations)